



GDCAU

Green Deal for Central Asian Universities

Project 101179245

How to Conduct a Simple Campus Sustainability Audit

A campus sustainability audit is a practical way to assess how “green” your college or university is and identify areas for improvement. It’s essentially a check-up on campus operations – from energy use and waste management to water conservation and transportation. This guide will walk you through a simple step-by-step process to conduct an audit, suitable for a coursework project or a student-led initiative. The focus is on keeping it straightforward and actionable, without requiring specialist expertise.

1. Set the Scope and Team

First, define *what* you will audit. It could be a broad audit covering multiple areas (e.g., energy, water, waste, etc.), or a focused audit on one aspect (like a **waste audit** in dorms, or an **energy audit** of one building). For beginners, a narrow scope might be easier – for example, auditing classroom electricity usage and recycling practices in one department. Form a small team if possible; sustainability audits are great as group projects. Divide roles: one person might handle data gathering (like utility bills), another does on-site observations, another takes notes and photos.

2. Gather Background Data

Before you physically inspect anything, collect any available data from campus facilities or administration. Useful data can include monthly electricity consumption, water usage records, waste collection volumes, etc. Even a few recent bills or reports can provide a baseline. For instance, knowing that your campus used **500,000 kWh of electricity last month** or **50 cubic meters of water per day** gives you a quantitative starting point. If such data is hard to get, don’t worry – the audit can still proceed based on observations and estimates, but any hard data you find will make your findings more concrete.

3. Use a Checklist for On-Site Inspection

Create a simple checklist of things to observe or count. Visit the areas relevant to your audit scope with this checklist in hand. Here are key categories and example checkpoints:

- **Energy Use:** Check classrooms, libraries, dorms, etc. Are lights and computers left on in empty rooms? How old are the light bulbs (are they LED or old incandescent bulbs)? Is natural sunlight being used where possible? You might do a “night walk” to see what’s left on after hours. Note things like the number of lights that could be turned off or opportunities for sensors/timers.
- **Waste Management:** Examine trash and recycling bins. Are recycling bins available and clearly labeled? Look inside a sample of bins – is there a lot of contamination (trash in recycling or vice versa)? You could perform a **small waste audit** by collecting one day’s worth of trash from a particular area and sorting it. For example, a dorm waste audit might reveal how much could have been recycled. In one campus event, students sorted a day’s trash from a residence hall and found significant amounts of recyclables thrown away incorrectly. That kind of data (e.g. “X% of trash was recyclables”) is powerful for raising awareness.
- **Water Usage:** Check restrooms, kitchens, labs for any leaks or drips (a dripping faucet can waste liters of water a day). Are there water-efficient fixtures (low-flow showerheads, dual-flush toilets) or not? If you have access to irrigation controls or schedules for campus landscaping, note if sprinklers run at optimal times (e.g. early morning to reduce evaporation). Even observational data like “we found 3 leaking taps in the science building” is valuable.
- **Transportation and Mobility:** Depending on your focus, you might survey how students and staff commute. Count bike racks (and how many bikes are on them), note if there are electric vehicle chargers or many cars idling. Perhaps do a quick tally at a parking lot: how full is it, any carpool or EV spots? Alternatively, a short survey of classmates (“How did you get to campus today?”) can give an idea of the transportation split (e.g. 50% public transit, 30% single-occupancy car, etc.).
- **Other Areas (if scope allows):** **Air and climate** (any monitoring on campus? Probably not for a simple audit, but if there are sensors check their data), **greenery and biodiversity** (count trees or green spaces, any campus gardens?), **procurement** (are cafeterias using disposable plastics or eco-friendly materials?), etc. You might not cover these in a basic audit, but keep an eye out for obvious issues (e.g. lots of single-use plastics in use).

Using a structured checklist ensures you don’t forget anything. You can make a table to tick off items, for example:

Area	What to Check	Observations
Classrooms (Energy)	Lights off when room empty? Equipment shutdown?	3 of 5 rooms had lights on during lunch with no occupants; old CRT monitor left on in one lab.

Area	What to Check	Observations
Dorm Waste	Recycling bins present and used?	Each floor has recycling bin, but found plastic bottles in trash (est. ~20% of trash could be recycled).
Restrooms (Water)	Leaks or running fixtures?	Identified 2 dripping faucets in Library restroom (constant drip).
Outdoor Lighting	LED vs older bulbs?	Most lamp posts are LED (good); however 5 pathway lights were on at noon (timer issue).

This kind of log will be the raw material for your audit report.

4. Analyze and Summarize Findings

After the walkthrough and data gathering, sit down with your notes and make sense of it. Look for patterns: maybe you found lights left on mainly in certain buildings, or a lack of recycling bins in student lounges. Quantify where you can (“roughly 30% of classroom lights were on unnecessarily”). Also, compare with any baseline data you got: e.g., if electricity use seems high, perhaps your observation of many appliances left on explains it. It’s often helpful to categorize findings into **Strengths** (what’s going well) and **Weaknesses** (areas to improve). For instance, *Strength*: “All new buildings have solar panels and LED lighting.” *Weakness*: “Older buildings rely on inefficient lighting and have no automatic controls.”

5. Research Solutions and Best Practices

For each weakness or issue you found, consider what could be done. This might involve a bit of quick research or common-sense ideas. If many recyclables are in the trash, perhaps the campus needs better signage or awareness campaigns about recycling. If lights are left on, maybe introduce motion sensors or a “lights out” campaign. If water leaks were found, a maintenance report should be filed to fix them (easy win). It can be helpful to see what other campuses have done. Many universities (including in Central Asia) have started green campus initiatives – for example, Nazarbayev University in Kazakhstan launched a “Green Campus” project that includes student-driven audits and energy-saving competitions. European universities often use frameworks like **STARS (Sustainability Tracking, Assessment & Rating System)** to guide their audits, which might inspire ideas.

6. Report and Share

Compile your findings and recommendations in a clear format. This could be a short report or presentation. Use visuals: photos you took during the audit (e.g. a photo of those dripping taps or lights on in an empty room) can be very convincing. Keep the tone positive and practical – the goal is to improve, not just to criticize. For each issue, suggest a solution and if

possible, an estimate of impact or cost. For example, “*Replacing 100 old bulbs with LEDs in Building X could cut power use by ~50% for lighting in that building, saving money and emissions.*” If your audit is for coursework, your instructor will value the real-world application; if it’s for the campus administration, they might actually implement some suggestions, especially if you show benefits like cost savings or enhanced student well-being.

7. Involve the Campus Community

A sustainability audit can double as an awareness activity. As you conduct it, talk to facility staff or fellow students – sometimes they’ll offer valuable insights (e.g. a janitor might tell you which recycling bins always get contaminated, or a lab manager might share that equipment X runs 24/7 for a reason). Involving people also softens resistance; if later you propose “Let’s turn off corridor lights at night,” those who heard about the audit are more likely to support it. Consider sharing fun facts or easy tips in a student newsletter or on social media: e.g. “*Our audit found 20 computers left on overnight – these could be adding 5 tons of CO₂ a year unnecessarily. Remember to shut down! #GreenCampus.*”

By following these steps, you conduct a thorough yet manageable campus sustainability audit. Even a simple audit can uncover opportunities to save resources and improve the campus environment. It’s hands-on learning at its best – you get to apply sustainability concepts directly to your daily surroundings. And who knows, your findings could kick-start a new green initiative at your university!

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November 2025*

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